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Inferring Dynamics of Discrete-time, Fractional-Order Control-affine Nonlinear Systems

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***Abstract*— In this paper, a system identification algorithm is proposed for control-affine, discrete-time fractional-order nonlinear systems. The proposed algorithm is a data-integrated framework that provides a mechanism for data generation and then uses this data to obtain the drift-vector, control-vector fields, and the fractional order of the system. The proposed algorithm includes two steps. In the first step, experiments are designed to generate the required data for dynamic inference. The second step utilizes the generated data in the first step to obtain the system dynamics. The memory-dependent property of fractional-order Grünwald-Letnikov difference operator is used to compute the fractional order of the system. Then, drift-vector and control-vector fields are reconstructed using orthonormal basis functions, and calculation of the coefficients is formulated as an optimization problem. Finally, simulation results are provided to illustrate the effectiveness of the proposed framework. Additionally, one of the methods for identifying integer-order systems is applied to the generated data by a fractional-order system, and the results are included to show the benefit of using a fractional-order model in long-range dependent processes.**

I. Introduction

Fractional Calculus is the generalization of integrals and derivatives to non-integer orders [1]. Fractional-order systems have been employed in control theory and dynamical systems thanks to their accuracy in modeling dynamical systems, their ability to achieve more stringent control requirements and to capture non-Markovian and Long-Range Dependences [2]. In the past few years, these systems have been used extensively to model systems, such as viscoelastic materials [3], diffusion wave [4], drug delivery [5], lithium-ion batteries [6], and power grids [7]. Currently, several control schemes have been developed for fractional-order systems, e.g., fractional PID controller [8] and fractional lead-lag controller [9]. Similarly, in order to improve the control performance of nonlinear systems, numerous control methods are developed based on fractional calculus, such as sliding mode control [10], [11], model reference adaptive control [12], and backstepping control [13].

Even though the introduction of fractional calculus in the theory of dynamical systems and signal processing demonstrates that models of non-Markovian and long-range dependence via fractional-order systems are more accurate than their integer-order counterparts [7], [14], inferring dynamics of these systems is still a challenging problem. In recent years, system identification methods have been developed for fractional-order systems. In [15], an identification method is proposed for fractional-order linear systems to estimate their parameters using time domain data based on the recursive least squares algorithm applied to an ARX structure derived from the fractional-order linear differential equation using adjustable fractional-order differentiators. Another method uses continuous order distributions to identify fractional-order systems [16]. In [17], fractional-order model parameters are computed by minimizing the sum of squared errors using a genetic algorithm. Moreover, a novel generalization of the Volterra LMS algorithm is proposed for fractional-order systems [18]. Finally, an iterative framework is proposed to determine the network parameterization and predict the state of discrete-time fractional dynamical systems [19].

To the best of our knowledge, we present the first data-integrated framework for dynamic inference of control-affine discrete-time fractional-order nonlinear systems. In the proposed algorithm, it is assumed that the system is control-affine and the number of state variables is known, but drift-vector, control-vector fields, and fractional order of the system are unknown. The proposed algorithm includes two main steps. In the first step, some experiments are designed to generate three datasets for dynamic inference. To generate the first dataset, a constant initial condition is considered with different control inputs are applied to the system so the state variables for the next two steps can be recorded. This experiment is then repeated for different initial conditions. The second and third datasets are generated following the same procedure with the difference that initial conditions are set to the system output in the first dataset. The second step of the algorithm involves finding the drift-vector, the control-vector fields, and the fractional order of the system. To achieve this, first, we use the memory-dependent property of fractional-order systems to calculate the fractional order of the system. Drift-vector and control-vector fields are approximated using orthonormal basis functions. Calculating the coefficients in the series is formulated as a supervised learning problem using mean squared error. Finally, to show the advantage of using fractional-order systems, one of the system identification methods is applied to the generated data and an integer-order system is obtained and time responses of fractional-order and integer-order systems are compared. Simulation results demonstrate that the memory dependence of the signal cannot be captured by integer-order systems while fractional-order systems can identify this property.

The rest of the paper is organized as follows: In Section II, preliminary definitions of fractional-order systems including Grünwald-Letnikov operator and discrete-time fractional-

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order systems are introduced. Section III defines the problem and the ultimate goal of the paper. In Section IV, the proposed algorithm is explained for dynamic inference in fractional-order nonlinear systems. In Section V, the proposed algorithm is used to identify dynamics of a fractional-order system and also an integer-order system is inferred using a previously-developed system identification method, and responses of two fractional-order and integer-order systems are compared to show the capability of fractional-order systems in modeling long-range dependent processes. The final section presents a brief conclusion.

II. Preliminary Concepts

The theory of non-integer order derivatives was first introduced by Leibniz [1]. After that, several definitions for fractional-order integrals, derivatives, and difference operators have been generated by mathematicians. In this section, some basic definitions of discrete-time fractional-order systems are introduced.

A. Grünwald-Letnikov Operator

The fractional-order Grünwald–Letnikov difference is defined as

$$\Delta^{\alpha} x_k \triangleq \sum_{j=0}^{k} \psi(\alpha, j) x_{k-j}, \tag{1}$$

where $x_k \in \mathbb{R}^n$, $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_n]^\top \in \mathbb{R}^n$ represents the order of the difference operator, and $\psi(\alpha, j) \in \mathbb{R}^{n \times n}$ is an $n \times n$ diagonal matrix defined as

$$\psi(\alpha, j) \triangleq \mathrm{diag}\big(\psi(\alpha_1, j), \psi(\alpha_2, j), \ldots, \psi(\alpha_n, j)\big), \tag{2}$$

with

$$\psi(\alpha_i, j) \triangleq \frac{\Gamma(j - \alpha_i)}{\Gamma(-\alpha_i)\Gamma(j+1)}, \quad i = 1, 2, \ldots, n, \tag{3}$$

where $\Gamma(\cdot)$ denotes the Gamma function defined as $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$.

B. Discrete-time Fractional-Order Systems

A discrete-time fractional-order nonlinear system in state-space representation is described as:

$$\Delta^{\alpha} x_{k+1} = h(x_k, u_k), \tag{4}$$

where $x_k \in \mathbb{R}^n$ is the system state vector, $u_k \in \mathbb{R}^m$ is the system input, $h(\cdot,\cdot) = \begin{bmatrix} h_1(\cdot,\cdot) & \cdots & h_n(\cdot,\cdot) \end{bmatrix}^\top$, and $h_i(\cdot,\cdot) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$, $i = 1, 2, \ldots, n$, are nonlinear functions. We assume that we can observe all the state variables of system (4).

Using the definition of Grünwald-Letnikov difference operator, one can rewrite system (4) as follows:

$$\begin{aligned} x_{k+1} &= \Delta^{\alpha} x_{k+1} - \sum_{j=1}^{k+1} \psi(\alpha, j) x_{k+1-j} \\ &= h(x_k, u_k) - \sum_{j=1}^{k+1} \psi(\alpha, j) x_{k+1-j}. \end{aligned} \tag{5}$$

The second term on the right-hand side of equation (5) shows the memory-dependent property of fractional-order dynamical systems. This property makes them appropriate for modeling non-Markovian and long-range dependent processes. In integer-order systems, this term is equal to zero, which means the next state in these systems depends only on the current state.

III. Problem Statement

In this paper, we focus on control-affine systems. Consider the following control-affine, discrete-time fractional-order nonlinear system:

$$\Delta^{\alpha} x_{k+1} = f(x_k) + g(x_k) u_k, \tag{6}$$

where $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_n]^\top \in \mathbb{R}^n$ represents the order of the difference operator, with $\alpha_i \in (0, 2)$, $i = 1, 2, \ldots, n$, $x_k \in \Omega \subseteq \mathbb{R}^n$ is the system state vector; $u_k \in \mathbb{R}^m$ is the system input, and $f(\cdot) : \Omega \to \mathbb{R}^n$ and $g(\cdot) : \Omega \to \mathbb{R}^{n \times m}$ are drift-vector and control-vector fields defined as

$$f(x_k) = \begin{bmatrix} f_1(x_k) & f_2(x_k) & \cdots & f_n(x_k) \end{bmatrix}^\top, \tag{7}$$

$$g(x_k) = \begin{bmatrix} g_{11}(x_k) & g_{12}(x_k) & \cdots & g_{1m}(x_k) \\ \vdots & \vdots & \ddots & \vdots \\ g_{n1}(x_k) & g_{n2}(x_k) & \cdots & g_{nm}(x_k) \end{bmatrix}, \tag{8}$$

where $f_i(\cdot) : \Omega \to \mathbb{R}$ and $g_{ij}(\cdot) : \Omega \to \mathbb{R}$, $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, m$, are nonlinear functions.

Following the same procedure that we used to obtain equation (5), We can rewrite system (6) in the following form:

$$x_{k+1} = f(x_k) + g(x_k) u_k - \sum_{j=1}^{k+1} \psi(\alpha, j) x_{k+1-j}. \tag{9}$$

Assumption 1: Throughout the paper, it is assumed that $f(\cdot)$ and $g(\cdot)$ are unknown analytical functions and fractional order of system (6) is unknown.

Problem 1: Consider system (6) with measurable state variables. Let the fractional order of the system, drift-vector, and control-vector fields be unknown. Find a system identification algorithm to calculate the fractional order, α, and functions $f_i(\cdot)$ and $g_{ij}(\cdot)$.

IV. Proposed Algorithm

In this section, a data-integrated framework is proposed for solving Problem 1. The proposed algorithm includes two main steps. In the first step, experiments are designed to generate the required datasets. The second step utilizes the generated data in the first step to formulate the dynamic inference as a supervised learning problem.

A. First Step: Data Generation

In this step, we design experiments to generate three datasets that we will use to infer the system dynamics. To achieve this, we consider a constant initial condition for system (9) and apply $(N+1)$ different control inputs, $q = 0, \ldots, N$, and record x_0, x_1, x_2, and x_3 for all the inputs. We repeat this experiment with M different initial conditions in set Ω, $p = 1, \ldots, M$. We generate the second dataset using x_1 and u_1 of the previous dataset as the initial condition and control input for the system, respectively. In the second set of experiments, we record only one step of the system response to calculate the order of the system. The last dataset is generated by setting the initial condition and control input to x_2 and u_2, respectively. In this dataset, we again record only one step of the system response.

Notation 1: x_0^p denotes the p^{th} initial condition in the above-mentioned experiments. u_0^{pq} and u_1^{pq} are the q^{th} control inputs for p^{th} initial condition in the first and second time steps, respectively. x_1^{pq} and x_2^{pq} stand for the output of the system after applying control inputs u_0^{pq} and u_1^{pq} with x_0^p initial condition. ${x'_2}^{pq}$ is the output of the system after applying control input u_1^{pq} with initial conditions x_1^{pq}. Morevoer, ${x'_3}^{pq}$ is the system response after applying control input u_2^{pq} with initial conditions x_2^{pq}.

B. Second Step: Dynamic Inference

In this step, first, we obtain the order of the system using the memory-dependent property of system (9) and then, we can find the control-vector field and subsequently the drift-vector field.

1) Fractional Order of the System: Since we only use two time steps of the system dynamics to generate the first dataset, we can simplify equation (9) for $k = 0$, $k = 1$, and $k = 2$ as follows:

$$\begin{aligned} x_1^{pq} &= f(x_0^p) + g(x_0^p)u_0^{pq} - \psi(\alpha,1)x_0^p \\ &= f(x_0^p) + g(x_0^p)u_0^{pq} + Ax_0^p, \end{aligned} \tag{10}$$

$$\begin{aligned} x_2^{pq} &= f(x_1^{pq}) + g(x_1^{pq})u_1^{pq} - \psi(\alpha,1)x_1^{pq} - \psi(\alpha,2)x_0^p \\ &= f(x_1^{pq}) + g(x_1^{pq})u_1^{pq} + Ax_1^{pq} + \frac{1}{2}\big(A - A^2\big)x_0^p, \end{aligned} \tag{11}$$

$$\begin{aligned} x_3^{pq} &= f(x_2^{pq}) + g(x_2^{pq})u_2^{pq} - \psi(\alpha,1)x_2^{pq} \\ &\quad - \psi(\alpha,2)x_1^{pq} - \psi(\alpha,3)x_0^p \\ &= f(x_2^{pq}) + g(x_2^{pq})u_2^{pq} + Ax_2^{pq} + \frac{1}{2}\big(A - A^2\big)x_1^{pq} \\ &\quad + \frac{1}{6}(A^3 - 3A^2 + 2A)x_0^p, \end{aligned} \tag{12}$$

where $A \in \mathbb{R}^{n\times n}$ is an $n \times n$ diagonal matrix with α_i, $i = 1, \ldots, n$, in its principal diagonal elements.

$$A = \operatorname{diag}\big(\alpha_1, \alpha_2, \ldots, \alpha_n\big). \tag{13}$$

By following the same procedure for the second and third datasets, one can obtain:

$$\begin{aligned} {x'_2}^{pq} &= f(x_1^{pq}) + g(x_1^{pq})u_1^{pq} - \psi(\alpha,1)x_1^{pq} \\ &= f(x_1^{pq}) + g(x_1^{pq})u_1^{pq} + Ax_1^{pq}, \end{aligned} \tag{14}$$

$$\begin{aligned} {x'_3}^{pq} &= f(x_2^{pq}) + g(x_2^{pq})u_2^{pq} - \psi(\alpha,1)x_2^{pq} \\ &= f(x_2^{pq}) + g(x_2^{pq})u_2^{pq} + Ax_2^{pq}. \end{aligned} \tag{15}$$

By subtracting equations (14) and (15) from (11) and (12), respectively, one can obtain:

$$x_2^{pq} - {x'_2}^{pq} = \frac{1}{2}\big(A - A^2\big)x_0^p, \tag{16}$$

$$x_3^{pq} - {x'_3}^{pq} = \frac{1}{2}\big(A - A^2\big)x_1^{pq} + \frac{1}{6}(A^3 - 3A^2 + 2A)x_0^p. \tag{17}$$

In (16), $(x_2^{pq} - {x'_2}^{pq})$ and x_0^p are known. Therefore, one can easily calculate the fractional order of the system by solving these algebraic equations. However, since these equations are quadratic, they have two real solutions. We choose the solutions which also satisfy (17).

Remark 1: The proposed algorithm for calculating the order of the system also holds for integer-order systems because the right-hand sides of (16) and (17) are equal to zero when the order is equal to one. In other words, an integer-order system has no memory.

Remark 2: While in the case where measurements are noiseless only four data points are needed, it is a better practice to use longer time series to achieve robustness against measurement noise.

2) Control-Vector Field: Let us define y^{pq} as

$$y^{pq} = x_1^{pq} - x_1^{p0}; \quad q = 1, \ldots, N, \quad p = 1, \ldots, M. \tag{18}$$

By substituting x_1^{pq} and x_1^{p0} from the system dynamics into (18), we can rewrite y^{pq} as

$$\begin{aligned} y^{pq} &= x_1^{pq} - x_1^{p0} \\ &= f(x_0^p) + g(x_0^p)u_0^{pq} + Ax_0^p - f(x_0^p) - g(x_0^p)u_0^{i0} - Ax_0^p \\ &= g(x_0^p)(u_0^{pq} - u_0^{p0}). \end{aligned} \tag{19}$$

In the last equation, $g(\cdot) \in \mathbb{R}^{n\times m}$ is an unknown matrix. We will approximate this function via a truncated series with orthonormal basis functions and calculate unknown coefficients via a least-square optimization problem.

Without loss of generality, we explain the method to $m = 1$. Extending the proposed method for $m > 1$ is straightforward.

Considering $m = 1$, the system input and control-vector field can be simplified as $u_k \in \mathbb{R}$ and $g(\cdot) = \big[g_1(\cdot) \quad \cdots \quad g_n(\cdot)\big]^\top$. We approximate g_i's with a truncated series using orthonormal basis functions of the underlying function space [20].

$$g_i(x_k) \approx \sum_{l=1}^{L} \beta_i^{(l)} \phi_i^{(l)}(x_k), \quad i = 1, \ldots, n, \tag{20}$$

where $\phi_i^{(l)}(\cdot)$ are orthonormal basis functions, $\beta_i^{(l)}$ are unknown coefficients, and L is the number of expansion terms in the truncated series.

Remark 3: Since we assumed that functions $f(\cdot)$ and $g(\cdot)$ are analytic, it is possible to find basis functions to approximate them.

By substituting equation (20) into (18), we obtain

$$y^{pq} = \begin{bmatrix} \sum_{l=1}^{L} \beta_1^{(l)} \phi_1^{(l)}(x_k) \\ \vdots \\ \sum_{l=1}^{L} \beta_n^{(l)} \phi_n^{(l)}(x_k) \end{bmatrix} \Delta u_0^{pq}, \tag{21}$$

where Δu_0^{pq} is the difference between u_0^{pq} and u_0^{p0}.

$$\Delta u_0^{pq} = u_0^{pq} - u_0^{p0}. \tag{22}$$

In equation (21), since we choose the orthonormal basis functions, the only unknown parameters in this equation are the series coefficients $\beta_i^{(l)}$. We can formulate the problem of calculating these coefficients as a least squares optimization problem. To do so, we define $X^{pq} \in \mathbb{R}^{(n)\times(nL)}$ and $\beta_i \in \mathbb{R}^L$ as

$$X^{pq} = \begin{bmatrix} \phi_1^{(1)}(x_0^p) & \cdots & \phi_1^{(L)}(x_0^p) & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \phi_2^{(1)}(x_0^p) & \cdots & \phi_2^{(L)}(x_0^p) & 0 & \cdots & 0 \\ \vdots & & \vdots & & & \ddots & & & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \phi_n^{(1)}(x_0^p) & \cdots & \phi_n^{(L)}(x_0^p) \end{bmatrix} \Delta u_0^{pq}, \tag{23}$$

$$\beta_i = \begin{bmatrix} \beta_i^{(1)} & \cdots & \beta_i^{(L)} \end{bmatrix}^\top, \quad i = 1, \dots, n. \tag{24}$$

We can stack all the data generated by the experiments and unknown parameters in the following form and formulate a regression problem. To achieve this, we define $Y \in \mathbb{R}^{MNn}$ as the label of the dataset, $X \in \mathbb{R}^{(MNn)\times(nL)}$ as the feature matrix, and $B \in \mathbb{R}^{nL}$ as unknown parameters. We define X, B, and Y as follows:

$$X = \begin{bmatrix} X^{11} \\ \vdots \\ X^{M1} \\ \vdots \\ X^{1N} \\ \vdots \\ X^{MN} \end{bmatrix}, \quad B = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}, \quad Y = \begin{bmatrix} y^{11} \\ \vdots \\ y^{M1} \\ \vdots \\ y^{1N} \\ \vdots \\ y^{MN} \end{bmatrix}. \tag{25}$$

Considering these matrices, we can rewrite the regression problem as follows:

$$Y = XB. \tag{26}$$

To calculate the unknown entries of B, we solve the following optimization problem.

$$B^* = \underset{B}{\arg\min} \;\; ||Y - XB||^2 \tag{27}$$

Using the normal equation, B can be obtained as

$$B^* = (X^\top X)^{-1}(X^\top Y). \tag{28}$$

Remark 4: Even though in the proposed algorithm we assume that $m = 1$, i.e. $g(\cdot) \in \mathbb{R}^n$ and $u_k \in \mathbb{R}$, extending the algorithm for $m > 1$ is straightforward. We only need to follow the same procedure for different entries of the system input vector while considering zero for other entries.

3) Drift-Vector Field: In order to find the drift-vector field, we only need equation (10). Since we have obtained the control-vector field and order of the system, the only unknown term in this equation is $f(x_0)$ which can be easily calculated by subtracting the left-hand side, x_1^{pq}, from all the terms in the right-hand side, $g(x_0)u_0 + Ax_0$. Note that we have chosen x_0 uniformly from set Ω to function $f(\cdot)$ in set Ω.

Remark 5: Optimization problem (27) is a linear regression problem with a convex cost function. Therefore, the solution is a global minimum and we can identify functions $g(\cdot)$ and $f(\cdot)$ and subsequently the system dynamics uniquely.

V. Simulation Results

In this section, the proposed algorithm is applied to a fractional-order nonlinear system to validate its effectiveness using simulation results. Furthermore, one of the algorithms for integer-order systems from the literature is employed to infer an integer-order system using the dataset generated by a fractional-order system. Responses of identified integer-order and fractional-order systems are compared to demonstrate the effectiveness of using fractional-order systems in long-range dependent processes.

A. Dynamic Inference of a fractional-order system

Consider the following discrete-time fractional-order logistic map.

$$\begin{aligned} \Delta^\alpha x_{k+1} = {} & \mu x_k(1 - x_k) \\ & + a\cos(x_k)\exp(b[\sin(x_k - c\pi) - 1])u_k. \end{aligned} \tag{29}$$

The well-known integer-order logistic map was proposed for generating a random number in the late 1940s. The fractional-order version of this map is studied extensively in the literature [21]. Note that by the nature of equations (25) and optimization problem (27), we can validate our theoretical results using scalar system (29) without loss of generality.

We use system (29) to generate the datasets we need for dynamic inference in the interval $x \in [0, 2\pi]$, with parameters $\alpha = 0.6$, $\mu = 1$, $a = -1$, $b = 4$, and $c = 0.7$.

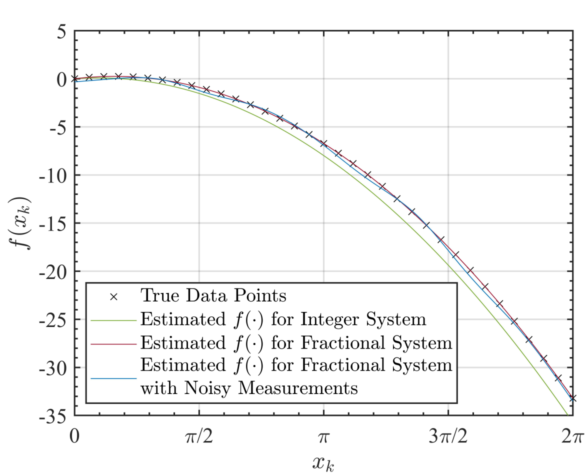

Fig. 1. True and Estimated Drift-Vector Fields.

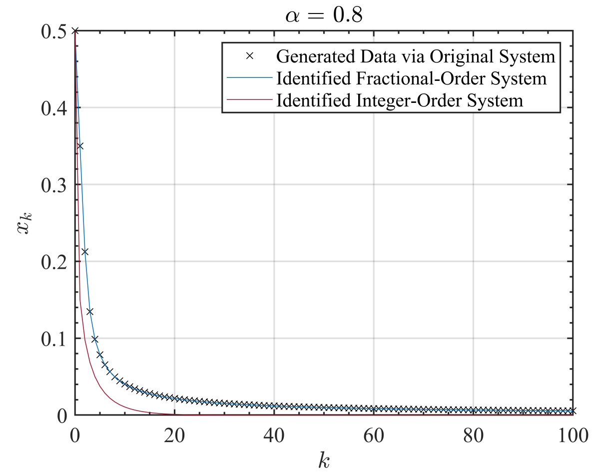

Fig. 2. System response for inferred fractional-order and integer-order systems with $\alpha = 0.8$.

Therefore, the drift-vector and control-vector fields are as

$$f(x_k) = x_k(1 - x_k), \tag{30}$$

$$g(x_k) = \exp(4[\cos(x_k - 0.7\pi) - 1]). \tag{31}$$

As mentioned in section IV, we approximate the control-vector field using the truncated Taylor series. In these simulations, we have considered $L = 7$ number of expansion terms in the truncated series. Simulation results for the drift-vector field of fractional-order and integer-order systems are shown in Figure 1. As seen in this figure, the proposed algorithm has approximated the true function accurately. The maximum absolute value of errors for estimated drift-vector fields of fractional-order and integer-order systems are 0.0038 and 1.2566, respectively. The drift-vector field is approximated perfectly. This happens because of using many data points in the interval of approximating this function and using a large number, i.e. $L = 7$ for expansion terms in truncated series. To evaluate robustness of the proposed algorithm against measurement noise, we approximated the drift-vector field using noisy observations, and the simulation result is shown in Figure 1.

B. Comparison of Fractional-Order Systems to Integer-Order Systems

In this subsection, we demonstrate the benefit of using fractional-order systems in modeling non-Markovian processes. To create a non-Markovian signal, we consider system (29) to generate the data. Then, we apply the proposed algorithm in this paper to identify a fractional-order system. Additionally, one of the data-driven system identification algorithms for integer-order systems introduced in [22] is employed to identify an integer-order system using the previously generated datasets.

In order to show the effect of the order of the system on system response and its memory dependence, simulation results are provided for two different orders, $\alpha = 0.8$ and $\alpha = 0.6$. System responses for identified fractional-order and integer-order systems are shown in Figures 2 and 3.

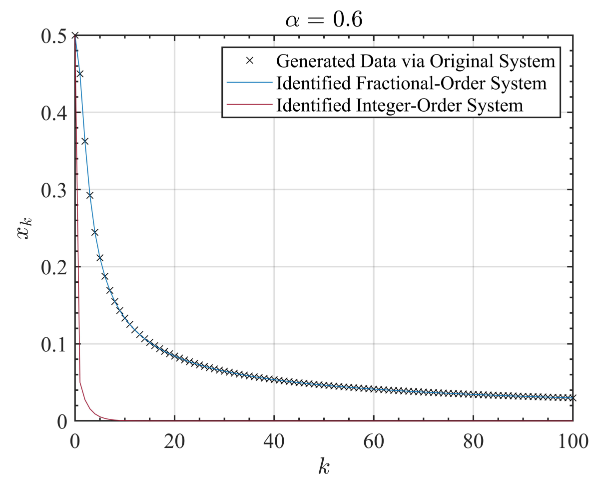

Fig. 3. System response for inferred fractional-order and integer-order systems with $\alpha = 0.6$.

While these simulations confirm the inability of integer-order systems to correctly identify the system, they also show how the error increases as α decreases, making the system more memory dependent, as shown in [23], [24].

VI. Conclusion

This paper proposes a data-integrated framework for identifying the dynamics of fractional-order systems. The proposed algorithm includes two main steps. The first step is designed to generate the required data to identify the system dynamics. The second step utilizes the generated data in the first step to approximate the drift-vector and control-vector fields. These functions are approximated via truncated series using orthonormal basis functions. The theoretical results are validated through numerical simulation results. The simulation results show that fractional-order systems model long-range dependent and non-Markovian processes more accurately because of their memory-dependent property.

Future research will study the effect of noisy measurements on the performance of the proposed algorithm and will focus on the design of a robust identification method using the Hurst exponent as a measure of the memory in a time series.

REFERENCES

[1] I. Podlubny, *Fractional differential equations: an introduction to fractional derivatives, fractional differential equations, to methods of their solution and some of their applications*. Elsevier, 1998.

[2] A. T. Azar, S. Vaidyanathan, and A. Ouannas, *Fractional order control and synchronization of chaotic systems*. Springer, 2017, vol. 688.

[3] F. Meral, T. Royston, and R. Magin, "Fractional calculus in viscoelasticity: an experimental study," *Communications in Nonlinear Science and Numerical Simulation*, vol. 15, no. 4, pp. 939–945, 2010.

[4] A. H. Bhrawy, E. H. Doha, D. Baleanu, and S. S. Ezz-Eldien, "A spectral tau algorithm based on jacobi operational matrix for numerical solution of time fractional diffusion-wave equations," *Journal of Computational Physics*, vol. 293, pp. 142–156, 2015.

[5] B. Yaghooti, M. Hosseinzadeh, and B. Sinopoli, "Constrained control of semilinear fractional-order systems: Application in drug delivery systems," in *4th IEEE Conference on Control Technology and Applications (CCTA)*. IEEE, 2020, pp. 833–838.

[6] Y. Wang, M. Li, and Z. Chen, "Experimental study of fractional-order models for lithium-ion battery and ultra-capacitor: Modeling, system identification, and validation," *Applied Energy*, vol. 278, p. 115736, 2020.

[7] L. Shalalfeh, P. Bogdan, and E. Jonckheere, "Evidence of long-range dependence in power grid," in *2016 IEEE Power and Energy Society General Meeting (PESGM)*. IEEE, 2016, pp. 1–5.

[8] B. Yaghooti and H. Salarieh, "Robust adaptive fractional order proportional integral derivative controller design for uncertain fractional order nonlinear systems using sliding mode control," *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, vol. 232, no. 5, pp. 550–557, 2018.

[9] H. K. Abdulkhader, J. Jacob, and A. T. Mathew, "Fractional-order lead-lag compensator-based multi-band power system stabiliser design using a hybrid dynamic ga-pso algorithm," *IET Generation, Transmission & Distribution*, vol. 12, no. 13, pp. 3248–3260, 2018.

[10] B. Yaghooti, A. Siahi Shadbad, K. Safavi, and H. Salarieh, "Adaptive synchronization of uncertain fractional-order chaotic systems using sliding mode control techniques," *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, vol. 234, no. 1, pp. 3–9, 2020.

[11] B. Yaghooti, K. Safavigerdini, R. Hajiloo, and H. Salarieh, "Stabilizing unstable periodic orbit of unknown fractional-order systems via adaptive delayed feedback control," *arXiv preprint arXiv:2208.06783*, 2022.

[12] H. Balaska, S. Ladaci, and A. Djouambi, "Direct fractional order mrac adaptive control design for a class of fractional order commensurate linear systems," *Journal of Control and Decision*, vol. 8, no. 3, pp. 363–371, 2021.

[13] F. Doostdar and H. Mojallali, "An adrc-based backstepping control design for a class of fractional-order systems," *ISA transactions*, vol. 121, pp. 140–146, 2022.

[14] M. Ghorbani, E. A. Jonckheere, and P. Bogdan, "Gene expression is not random: scaling, long-range cross-dependence, and fractal characteristics of gene regulatory networks," *Frontiers in physiology*, vol. 9, p. 1446, 2018.

[15] D. Idiou, A. Charef, and A. Djouambi, "Linear fractional order system identification using adjustable fractional order differentiator," *IET Signal Processing*, vol. 8, no. 4, pp. 398–409, 2014.

[16] T. T. Hartley and C. F. Lorenzo, "Fractional-order system identification based on continuous order-distributions," *Signal processing*, vol. 83, no. 11, pp. 2287–2300, 2003.

[17] P. Shah and R. Sekhar, "Closed loop system identification of a dc motor using fractional order model," in *2019 International Conference on Mechatronics, Robotics and Systems Engineering (MoRSE)*. IEEE, 2019, pp. 69–74.

[18] N. I. Chaudhary, M. A. Z. Raja, M. S. Aslam, and N. Ahmed, "Novel generalization of volterra lms algorithm to fractional order with application to system identification," *Neural Computing and Applications*, vol. 29, no. 6, pp. 41–58, 2018.

[19] G. Gupta, S. Pequito, and P. Bogdan, "Learning latent fractional dynamics with unknown unknowns," in *2019 American Control Conference (ACC)*. IEEE, 2019, pp. 217–222.

[20] H. Akçay and B. Ninness, "Orthonormal basis functions for modelling continuous-time systems," *Signal processing*, vol. 77, no. 3, pp. 261–274, 1999.

[21] J. Munkhammar, "Chaos in a fractional order logistic map," *Fractional Calculus and Applied Analysis*, vol. 16, no. 3, pp. 511–519, 2013.

[22] V. Narayanan, W. Miao, and J.-S. Li, "Disentangling drift-and control-vector fields for interpretable inference of control-affine systems," *arXiv preprint arXiv:2004.10954*, 2020.

[23] G. Gupta, C. Yin, J. V. Deshmukh, and P. Bogdan, "Non-markovian reinforcement learning using fractional dynamics," in *2021 60th IEEE Conference on Decision and Control (CDC)*. IEEE, 2021, pp. 1542–1547.

[24] P. Flandrin, "Wavelet analysis and synthesis of fractional brownian motion," *IEEE Transactions on information theory*, vol. 38, no. 2, pp. 910–917, 1992.